

Chapter 1~2

For each of the following questions there are four choices marked A, B, C and D. Only one of them is correct. You should decide on the correct choice and write down its mark on the answer sheet.

1. We can identify five major components in the DBMS environment. They are hardware, software, procedures, people and C.
A. database B. DBMS C. data D. DBS
2. The relationship among DB, DB application and DBS is A.
A. A DB system contains multiple DB applications, which access DB
B. A DB application contains only one DB system, which accesses DB
C. A DB contains multiple DB applications, which access DB system
D. A DB application contains multiple DBs, which access DB system
3. The most appropriate method to manage data with high data sharing and low data redundancy is D. 高数据共享 低数据冗余
A. file-based approach B. entity-relationship approach 整体表示
C. data-driven approach 数据分割 D. database approach
4. In the following properties, the one that does not belong to database approach is C.
A. high data sharing B. improved data integrity
C. high data redundancy D. improved data independence
5. In database systems, data independence means that C. immunity 免疫
A. the immunity of application programs to changes of the data value 值
B. the immunity of application programs to changes of the data amounts 数量
C. the immunity of application programs to changes of the data structure 结构
D. the immunity of the data storage structure to changes of the data storage approach
6. The person who manages the data resource involved in DB is referred to as D.
A. DBA B. EMBA C. MBA D. DA
7. The person who is responsible for the physical realization of the database is referred to as A.
A. DBA B. EMBA C. MBA D. DA
8. A contains the global logical representation of the entire database.
A. Conceptual schema B. Internal schema 物理逻辑表示
C. External schema D. Physical schema
9. It is C that ensures the data independence in database system.
A. the schema and the external schema
B. the schema and the internal schema
C. the two mappings between the three level of schemas 三层模式的两个映射
D. the three level of schemas
10. It is A that ensures the logical data independence in database system.
A. the mappings between conceptual schema and external schemas
B. the mapping between conceptual schema and internal schema
C. the conceptual schema
D. the three level of schemas
11. The description of the database logical structure that can be seen by both (users) and

(application programs) is called C.

- A. conceptual schema
 - B. internal schema
 - C. external schema
 - D. physical schema
12. Which of the following tasks is not a duty of DBA ? D
- A. defining integrity constraints
 - B. defining database schema
 - C. managing database security
 - D. designing database management system
13. Which of the following operations is not supported by DML ?
- A. inserting some rows into a table
 - B. deleting some rows from a table
 - C. finding some rows from a table
 - D. creating some tables in a database
14. Which of the following operations is not supported by DDL ?
- A. creating some views in a database
 - B. modifying table structures in a database
 - C. finding some rows from a table
 - D. creating some tables in a database
15. Among object-based data models, the most famous one is _____.
- A. entity-relationship data model
 - B. semantic data model
 - C. relational data model
 - D. functional data model
16. A data model comprises three components, they are _____.
- A. the external schema, the conceptual schema and the internal schema
 - B. the relations, the tuples and the primary keys
 - C. the structural part, the manipulative part and the integrity rules
 - D. the meaning of model, the operational rules and the implementation language
17. The data in a hierarchical data model is of _____.
- A. a tree structure
 - B. a directed graph structure
 - C. a linear structure
 - D. a tabular structure
18. The data in a network data model is of _____.
- A. a tree structure
 - B. a directed graph structure
 - C. a linear structure
 - D. a tabular structure
19. A relational data model is the most important model that is composed of _____.
- A. the relational data structure, the relational operations and the integrity constraints
 - B. the relations, the tuples and the primary keys
 - C. the entity integrity, the referential integrity and the user-defined integrity
 - D. the external schema, the conceptual schema and the internal schema
20. If we query in a database based on a _____, we need only to express *what* and not *how*.
- A. hierarchical data model
 - B. network data model
 - C. relational data model
 - D. physical data model
21. Which of the following is true in database systems?
- A. DBMS runs concurrently with OS.
 - B. DBMS and OS call each other.
 - C. OS calls DBMS.
 - D. DBMS calls OS.
22. In designing file-based systems, as the main problem is the design of system functions, among the following works, the most important one is _____.
- A. designing data structure
 - B. programming
 - C. memory arrangement
 - D. information storage
23. The core of the information processing based on database systems is _____.
- A. the disk
 - B. program
 - C. the data
 - D. the memory
24. In database system, the conversion between logical data and physical data can be done

- by ____.
- A. application programs B. the OS C. the DBMS D. I/O facilities
25. There is _____ among the three-level schemas of a database system.
- A. integrity B. consistency C. structural consistency D. a great difference
26. The three-level architecture of database systems is the three-level abstractions for ____.
- A. memories B. data C. programs D. disks
27. Among the three-level schemas of database systems, the nearest one from the secondary storage is ____.
- A. the external schemas B. the conceptual schema
C. the subschemas D. the internal schema
28. It is just because _____ that there is data independence in database systems.
- A. the disk-based storage B. the three-level schemas
C. the data access by calling OS D. the visualized programming language
29. In database systems, the relation between 'data independence' and 'data relationship' is that ____.
- A. they are irrelevant concepts B. they are simultaneously true or false
C. the former infers the latter D. they can infer each other
30. In database system, 'data independence' means that ____.
- A. this data is independent on that data
B. application programs are independent on the database structure
C. disk is independent on memory
D. data is independent on disk

Chapter 3~4

For each of the following questions there are four choices marked A, B, C and D. Only one of them is correct. You should decide on the correct choice and write down its mark on the answer sheet.

- Which one of the following is not the property of relations?

A. Each attribute has a distinct data type. B. Each attribute has a distinct name.
C. Order of rows has no significance. D. Order of cols has no significance.
- In relational models, each relation can only have one _____.

A. candidate key B. foreign key C. inner key D. primary key
- In relational models, each candidate key _____.

A. can only have at least two attributes B. can only have at most one attributes
C. may have one or more attributes D. must contain all the attributes of the relation
- Given a relation schema R and its attributes X , if X references a candidate key Y of relation schema S , then we call _____.

A. X is a foreign key of relation schema S B. Y is a foreign key of relation schema R
C. X is a foreign key of relation schema R D. Y is a foreign key of relation schema S
- Suppose Set D_1 , D_2 and D_3 have K_1 , K_2 and K_3 elements, respectively, then the number of tuples in $D_1 \times D_2 \times D_3$ will be _____.

- A. $K_1 \times K_2 \times K_3$ B. $K_1 + K_2 + K_3$ C. $(K_1 + K_2) \times K_3$ D. $(K_1 + K_2) \div K_3$
6. Special relational operations that a relational DBMS should support include: _____.
 - A. sorting, indexing and statistical operation
 - B. selection, projection and join
 - C. association, update and sorting
 - D. display, print, tabulation
 7. Between relation schema and relational database schema, which of the following is true?
 - A. Relation schema is a type, relational database schema is its value.
 - B. Relational database schema is a type, relation schema is its value.
 - C. Relation schema is a set of relational database schemas.
 - D. Relational database schema is a set of relation schemas.
 8. Between relation and relation schema, which of the following is true?
 - A. Relation schema is a type, relation is its value.
 - B. Relation is a type, relation schema is its value.
 - C. Relation schema is a set of relations.
 - D. Relation is a set of relation schemas.
 9. Between relational database and relational database schema, which of the following is true?
 - A. Relational database is a type, relational database schema is its value.
 - B. Relational database schema is a type, relational database is its value.
 - C. Relational database schema is a set of relational databases.
 - D. Relational database is a set of relational database schemas.
 10. Between relation and relational database, which of the following is true?
 - A. Relational database is a type, relation is its value.
 - B. Relation is a type, relational database is its value.
 - C. Relational database is a set of relations.
 - D. Relation is a set of relational databases.
 11. Null represents a value for an attribute in a tuple that is _____.
 - A. zero
 - B. an empty set
 - C. a space
 - D. currently unknown or not applicable
 12. Relation schema is _____.
 - A. the association schema between two relations
 - B. the association schema between a relation and a tuple
 - C. the description of a relation
 - D. the description of the relationship between two relations
 13. If F is a foreign key of relation R that targets a candidate key K of relation S , then F can only take _____.
 - A. null values
 - B. null values or values that appear in K
 - C. values that appear in K
 - D. values that appear in R
 14. Relational integrity means _____.
 - A. the correctness and consistency of data in relations
 - B. that every relation must integrally contain all the possible data
 - C. that every data must be integrally represented in relations
 - D. that we must integrally describe specific problems using relations
 15. In relational model, entity integrity means that _____.
 - A. there must not be empty rows in every relation
 - B. no attribute of a primary key can be null
 - C. attributes are allowed to be null

- D. foreign keys are allowed to be null
16. The main property of relational operations in relational algebra is that _____.
 A. both the operational objects and the result are fields
 B. the operational objects are fields, but the result is a relation
 C. the operational objects are relations, but the result is a field
 D. both the operational objects and the result are relations
17. Suppose R and S are n -ary relations with the same attribute set, then the difference of R and S , denoted as _____, is composed of tuples that belong to R but not to S .
 A. $R - S$ B. $S - R$ C. $R \cup S$ D. $S \cap S$
18. The projection operation in relational database refers to _____ a relation.
 A. the extraction of specific records from B. the extraction of specific fields from
 C. the creation of specific images of D. the creation of specific graphs of
19. Which of the following operations can extract from a relation all records that satisfy a condition to form a new relation?
 A. Projection B. Join C. Selection D. Division
20. In general, to conduct natural join between relation R and S , it is necessary for R and S to possess common _____.
 A. records B. rows C. attributes D. tuples
21. The degree of a relation refers to _____ in the relation.
 A. the number of rows B. the number of tuples
 C. the number of candidate keys D. the number of columns
22. Suppose there are 10 tuples in each of relation R , S and W , then the number of tuples in $R \bowtie S \bowtie W$ is _____.
 A. 10 B. 30 C. 1000 D. unknown
23. Suppose there are respectively 2 and 3 attributes in relation R and S , then $R \bowtie_{1<2} S$ is equivalent to _____.
 A. $\sigma_{1<2}(R \times S)$ B. $\sigma_{1<4}(R \times S)$ C. $\sigma_{1<2}(R \bowtie S)$ D. $\sigma_{1<4}(R \bowtie S)$
24. If there is no common attributes in two relations, then the natural join between them is equivalent to _____.
 A. Cartesian product B. equi-join C. θ -join D. outer join
25. Which of the following equations is wrong?
 A. $R - S = R - (R \cap S)$ B. $R = (R - S) \cup (R \cap S)$
 C. $R \cap S = S - (S - R)$ D. $R \cap S = S - (R - S)$
26. Suppose we are given relation $R(A, B, C)$ and $S(B, C, D)$, then which of the following relational algebra expressions is equivalent to $R \bowtie S$?
 A. $\sigma_{3=5}(R \bowtie_{2=4} S)$ B. $\pi_{1,2,3,6}(\sigma_{3=5}(R \bowtie_{2=4} S))$
 C. $\sigma_{3=5 \wedge 2=4}(R \times S)$ D. $\pi_{1,2,3,6}(\sigma_{3=2 \wedge 2=1}(R \times S))$
27. Suppose we are given relation $R(A, B, C)$ and $S(B, C, D)$, then which of the following relational algebra expressions is equivalent to $R \bowtie S$?
 A. $\sigma_{3=5}(R \bowtie_{2=4} S)$ B. $\pi_{1,2,3,6}(\sigma_{3=5 \wedge 2=4}(R \times S))$
 C. $\sigma_{3=5 \wedge 2=4}(R \times S)$ D. $\pi_{1,2,3,6}(\sigma_{3=2 \wedge 2=1}(R \times S))$

28. Suppose we are given relation $R(A,B,C)$ and $S(B,C)$, then which of the following relational algebra expressions is equivalent to $R \bowtie S$?
- $\{a \mid a \in \Pi_A(R) \wedge \Pi_B(\sigma_{A=a}(R)) \supseteq \Pi_B(S)\}$
 - $\{a \mid a \in \Pi_A(R) \wedge \Pi_{B,C}(\sigma_{A=a}(R)) \supseteq S\}$
 - $\{b \mid b \in \Pi_B(S) \wedge \Pi_B(\sigma_{B=b}(S)) \supseteq \Pi_B(R)\}$
 - $\{b \mid b \in \Pi_B(S) \wedge \sigma_{B=b}(S) \supseteq \Pi_{B,C}(R)\}$

Chapter 5~6

For each of the following questions there are four choices marked A, B, C and D. Only one of them is correct. You should decide on the correct choice and write down its mark on the answer sheet.

- SQL is _____ language.
 - a procedural
 - a host
 - an assemble
 - a non-procedural
- Which of the following operations cannot be carried out using SQL?
 - updating views
 - creating base tables based on views
 - querying views
 - creating new views based on old views
- In SQL, CREATE, DROP and ALTER are _____ commands.
 - data query
 - data modification
 - data definition
 - data control
- In SQL, in order to delete a view, we use _____ command.
 - DELETE
 - DROP
 - CLEAR
 - REMOVE
- In SQL, views are corresponding to the _____ schema of a database.
 - external
 - conceptual
 - inner
 - storage
- Which of the following SQL statements is not a data definition statement?
 - CREATE TABLE
 - DROP VIEW
 - CREATE VIEW
 - GRANT
- In SQL, in order to delete an existing base table S, we use _____ command.
 - DELETE TABLE S
 - DELETE S
 - DROP TABLE S
 - DROP S
- In SQL, in order to add a new column CN into base table S, we use _____ command.
 - ADD TABLE S(CN CHAR(8))
 - ADD TABLE S ALTER(CN CHAR(8))
 - ALTER TABLE S ADD CN CHAR(8)
 - ALTER TABLE S (ADD CN CHAR(8))
- Given table $S(Sno, Sname, Ssex, Sage)$, in order to delete column Sage of S using SQL, we shall use _____ command.
 - DELETE Sage FROM S
 - ALTER TABLE S DROP Sage
 - UPDATE S Sage
 - ALTER TABLE S ' Sage'
- Suppose we are given relation $S(Sno \text{ CHAR}(5), Sname \text{ CHAR}(10), Sage \text{ INT})$, $SC(Sno \text{ CHAR}(5), Cno \text{ CHAR}(5), Grade \text{ FLOAT})$ and $C(Cno \text{ CHAR}(5), Cname \text{ CHAR}(40))$, where a row in S means that a student with student number *Sno* and name *Sname* has age *Sage*, a row in SC means that a student with student number *Sno* takes part in a course with course number *Cno* and obtains a grade *Grade*, and a row in C means that a course with course number *Cno* has course name *Cname*, then in order to list student name of each student that takes part in course 'ACCESS' and has a age not less than 20, the possible SQL statement we may use will be SELECT SNAME FROM S, C, SC WHERE _____.

- A. S.Sno=SC.Sno AND C.Cno=SC.Cno AND Sage >=20 AND Cname = ' ACCESS'
 - B. S.Sno=SC.Sno AND C.Cno=SC.Cno AND Sage >=20 AND Cname IN ' ACCESS'
 - C. Sage >=20 AND Cname IN ' ACCESS'
 - D. Sage >=20 AND Cname = ' ACCESS'
11. Suppose we are given relation $S(SN \text{ CHAR}(10), CN \text{ CHAR}(40), GRADE \text{ FLOAT})$, where a row in S means that a student with student name SN takes part in a course with course name CN and obtains a grade $GRADE$, then in order to insert information " Student ' Zhangming' takes part in course ' database' and obtains a grade 80" into S , the possible SQL statement we may use will be ____.
- A. ADD INTO S VALUES(' Zhangming' , ' database' , ' 80')
 - B. INSERT INTO S VALUES(' Zhangming' , ' database' , ' 80')
 - C. ADD INTO S VALUES(' Zhangming' , ' database' , 80)
 - D. INSERT INTO S VALUES(' Zhangming' , ' database' , 80)
12. Suppose we are given relation $S(SN \text{ CHAR}(10), CN \text{ CHAR}(40), GRADE \text{ FLOAT})$, where a row in S means that a student with student name SN takes part in a course with course name CN and obtains a grade $GRADE$, then in order to set Zhangming' s database grade to 85, the possible SQL statement we may use will be ____.
- A. UPDATE S SET GRADE=85 WHERE SN=' Zhangming' and CN=' database'
 - B. UPDATES SET GRADE=' 85' WHERE SN=' Zhangming' and CN=' database'
 - C. UPDATE GRADE=85 WHERE SN=' Zhangming' and CN=' database'
 - D. UPDATE GRADE=' 85' WHERE SN=' Zhangming' and CN=' database'
13. In SQL, subquery is a query statement that ____.
- A. returns a subset of data in one table
 - B. returns a subset of fields in multiple tables
 - C. selects a subset of fields in one table
 - D. is embedded into another query statement
14. Suppose we are given relation $S(Sno \text{ CHAR}(5), Sname \text{ CHAR}(10), Ssex \text{ INT})$, $SC(Sno \text{ CHAR}(5), Cno \text{ CHAR}(5), Grade \text{ FLOAT})$ and $C(Cno \text{ CHAR}(5), Cname \text{ CHAR}(40))$, where a row in S means that a student with student number Sno and name $Sname$ has sex $Ssex$, a row in SC means that a student with student number Sno takes part in a course with course number Cno and obtains a grade $Grade$, and a row in C means that a course with course number Cno has course name $Cname$, then in order to list student name of each male student that takes part in course ' database' , the possible SQL statement we may use will be SELECT SNAME FROM S, C, SC WHERE ____.
- A. S.Sno=SC.Sno and C.Cno=SC.Cno and Ssex=' M' and Cname = ' database'
 - B. S.Sno=SC.Sno and C.Cno=SC.Cno and Ssex in ' M' and Cname in ' database'
 - C. Ssex IS ' M' and Cname IS ' database'
 - D. Ssex=' M' and Cname = ' database'
15. If we create a table SC using the following SQL statement:
- ```
CREATE TABLE SC(Sno CHAR(6) NOT NULL,
 Cno CHAR(3) NOT NULL,
 Grade INT,
 NOTE CHAR(20)),
```
- then SQL statement INSERT INTO SC VALUES \_\_\_\_ can be executed successfully.
- A. ( ' 201009' , ' 111' , 60, compulsory)
  - B. ( ' 200823' , ' 101' , NULL, NULL)

- C. (NULL, ' 103' , 80, ' compulsory ' )      D. ( ' 201132' , NULL, 86, ' ' )
16. Suppose we are given relation  $S(Sno\ CHAR(5), Sname\ CHAR(10), Ssex\ INT)$ ,  $SC(Sno\ CHAR(5), Cno\ CHAR(5), Grade\ FLOAT)$  and  $C(Cno\ CHAR(5), Cname\ CHAR(40))$ , where a row in  $S$  means that a student with student number  $Sno$  and name  $Sname$  has sex  $Ssex$ , a row in  $SC$  means that a student with student number  $Sno$  takes part in a course with course number  $Cno$  and obtains a grade  $Grade$ , and a row in  $C$  means that a course with course number  $Cno$  has course name  $Cname$ , then in order to list student name of each male student that takes part in course ' database' , tables must be involved are \_\_\_\_.
- A. S, C      B. S, SC      C. C, SC      D. S, C, SC
17. In SQL, SELECT clause can realize \_\_\_\_ operation of relational algebra. A
- A. projection      B. selection      C. join      D. intersection
18. In SQL, WHERE clause can realize \_\_\_\_ operation of relational algebra.
- A. projection      B. selection      C. join      D. intersection
19. In SQL, in order to eliminate repeated rows in query results, which of the following key word should be used?
- A. UNIQUE      B. COUNT      C. DISTINCT      D. UNION
20. Which of the following SQL operation is not correct (where AGE is a field name of a table)?
- A. AGE IS NULL      B. AGE IS NOT NULL  
C. AGE=NULL      D. NOT ( AGE IS NULL )
21. Suppose we are given relation  $DEPT(DNO\ CHAR(5), DNAME\ CHAR(20))$ , in order to find the DNAMEs that contain a least 4 characters and if count backwards, the 3<sup>rd</sup> alphabet of which is W, the WHERE clause may be WHERE DNAME LIKE \_\_\_\_.
- A. '\_ \_W \_%'      B. '\_ W \_ %'      C. '\_ W \_ \_'      D. '\_ %W \_ \_'
22. If there are multiple tables involved in a SELECT statement, then the WHERE clause of the statement must include \_\_\_\_.
- A. only selection conditions      B. joining conditions or selection conditions  
C. only joining conditions      D. joining conditions and selection conditions
23. In SQL, the operation that is equivalent to NOT IN is \_\_\_\_.
- A. = ANY      B. <> ANY      C. = ALL      D. <> ALL
24. In SQL, the operation that is equivalent to SALARY IN (1000, 2000) is \_\_\_\_.
- A. SALARY <=2000 AND SALARY >=1000      B. SALARY <2000 AND SALARY >1000  
C. SALARY =2000 AND SALARY=1000      D. SALARY =2000 OR SALARY=1000
25. The execution order of clauses in a SELECT statement is \_\_\_\_.
- A. 1)SELECT; 2)FROM; 3)WHERE; 4)GROUP BY; 5)HAVING; 6)ORDER BY  
B. 1)WHERE; 2)FROM; 3)SELECT; 4)GROUP BY; 5)HAVING; 6)ORDER BY  
C. 1)FROM; 2)WHERE; 3)SELECT 4)GROUP BY; 5)HAVING; 6)ORDER BY  
D. 1)FROM; 2)WHERE; 3)GROUP BY; 4)HAVING; 5)SELECT 6)ORDER BY
26. Suppose we are given relation  $C(Cno\ CHAR(5), Cname\ CHAR(40))$  and  $SC(Sno\ CHAR(5), Cno\ CHAR(5), Grade\ FLOAT)$ , then which one of the following SQL statements has no grammar error?
- A. SELECT C.Cno, Cname, AVG(Grade) FROM C, SC  
WHERE C.Cno=SC.Cno  
GROUP BY Cno, Cname HAVING AVG(Grade)>=60  
B. SELECT Cno, Cname, AVG(Grade) FROM C, SC

- WHERE C.Cno=SC.Cno AND AVG(Grade)>=60
- C. SELECT C.Cno, Cname, AVG(Grade) FROM C, SC  
WHERE C.Cno=SC.Cno  
GROUP BY C.Cno,Cname HAVING AVG(Grade)>=60
- D. SELECT C.Cno, Cname, AVG(Grade) FROM C, SC  
WHERE C.Cno=SC.Cno AND AVG(Grade)>=60
27. Suppose we are given relation  $S(Sno\ CHAR(5), Sname\ CHAR(40))$  and  $SC(Sno\ CHAR(5), Cno\ CHAR(5), Grade\ FLOAT)$ , then which one of the following SQL statements has no grammar error?
- A. SELECT S.Sno, Sname, AVG(Grade) FROM S, SC  
WHERE S.Sno=SC.Sno  
GROUP BY Sno, Sname HAVING AVG(Grade)>=60
- B. SELECT Sno, Sname, AVG(Grade) FROM S, SC  
WHERE S.Sno=SC.Sno AND AVG(Grade)>=60
- C. SELECT S.Sno, Sname, AVG(Grade) FROM S, SC  
WHERE S.Sno=SC.Sno  
GROUP BY S.Sno, Sname HAVING AVG(Grade)>=60
- D. SELECT S.Sno, Sname, AVG(Grade) FROM S, SC  
WHERE S.Sno=SC.Sno AND AVG(Grade)>=60

## Chapter 13

For each of the following questions there are four choices marked A, B, C and D. Only one of them is correct. You should decide on the correct choice and write down its mark on the answer sheet.

- In relational database, each attribute of a relation must be \_\_\_\_\_.  
A. invariable      B. indivisible      C. reliable      D. available
- Given relation schema  $R(A,B,C,D,E)$  and its FD set  $F=\{A \rightarrow D, B \rightarrow C, E \rightarrow A\}$ , then the candidate key of R is \_\_\_\_\_.  
A. AB      B. BE      C. CD      D. DE
- If a relation schema belongs to 2NF, then it \_\_\_\_\_.  
A. may belong to 1NF      B. certainly belongs to 1NF  
C. certainly belongs to 3NF      D. certainly belongs to BCNF
- If all attributes of relation schema  $R$  are primary attributes, then  $R$  at least belongs to \_\_\_\_\_.  
A. 1NF      B. 2NF      C. 3NF      D. BCNF
- If there is no partial FD in relation schema  $R$  that belongs to 1NF, then  $R$  certainly belongs to \_\_\_\_\_.  
A. 1NF      B. 2NF      C. 3NF      D. BCNF
- A relation schema \_\_\_\_\_.  
A. can have exactly one candidate key      B. can have many primary keys  
C. can have one or many candidate keys      D. can have one or many primary keys
- A candidate key can contain \_\_\_\_\_.  
A. zero or many attributes      B. no attribute  
C. exactly one attribute      D. one or many attributes
- Suppose the decomposition of relation schema  $R(A,B,C)$  is  $\rho=\{AB, AC\}$ , then when  $R'$ 's FD set  $F$  equals \_\_\_\_\_,  $\rho$  will be a lossless-join decomposition.  
A.  $\{B \rightarrow C\}$       B.  $\{C \rightarrow B\}$       C.  $\{A \rightarrow C\}$       D.  $\{C \rightarrow A\}$

9. Suppose the decomposition of relation schema  $R(A,B,C)$  is  $\rho = \{ AB, AC \}$ , then when  $R'$ 's FD set  $F$  equals \_\_\_\_\_,  $\rho$  will be a lossless-join and dependency-preserving decomposition.
- A.  $\{ A \rightarrow B \}$       B.  $\{ A \rightarrow B, B \rightarrow C \}$       C.  $\{ B \rightarrow A \}$       D.  $\{ C \rightarrow B, B \rightarrow A \}$
10. Given relation schema  $R(S, D, M)$  and its FD set  $F = \{ S \rightarrow D, D \rightarrow M \}$ , then at most  $R$  belongs to \_\_\_\_\_.
- A. 1NF      B. 2NF      C. 3NF      D. BCNF
11. Given relation schema  $R(A,B,C,D)$  and its FD set  $F = \{ AB \rightarrow C, C \rightarrow D \}$ , then at most  $R$  belongs to \_\_\_\_\_.
- A. 1NF      B. 2NF      C. 3NF      D. BCNF
12. If \_\_\_\_\_, then we call  $X \rightarrow Y$  trivial FD.
- A.  $X \subseteq Y$       B.  $Y \subseteq X$       C.  $X \cap Y = \phi$       D.  $X \cap Y \neq \phi$
13. In relation schema  $R$ ,  $X \rightarrow Y$  means that \_\_\_\_\_.
- A. In one of  $R'$ 's relation, if there are two rows that equal on  $X$ , then they also equal on  $Y$ .
- B. In each of  $R'$ 's relation, if there are two rows that equal on  $X$ , then they also equal on  $Y$ .
- C. In one of  $R'$ 's relation, values of  $Y$  equal values of  $X$ .
- D. In each of  $R'$ 's relation, values of  $Y$  equal values of  $X$ .
14. \_\_\_\_\_ means that If  $X \rightarrow Y$  and  $WY \rightarrow Z$ , then  $WX \rightarrow Z$ .
- A. Augmentation      B. Transitivity      C. Pseudo-transitivity      D. Decomposition
15.  $X \rightarrow Y$  can be inferred by Armstrong's axioms if and only if \_\_\_\_\_.
- A.  $Y \subseteq X$       B.  $Y \subseteq X^+$       C.  $X \subseteq Y^+$       D.  $X^+ = Y^+$
16. For FD sets  $F$  and  $G$ , If \_\_\_\_\_, then we say  $F$  and  $G$  are equivalent to each other.
- A. every FD that can be inferred by  $F$  can also be inferred by  $G$
- B. every FD that cannot be inferred by  $F$  also cannot be inferred by  $G$
- C.  $F = G$
- D. every FD that can be inferred by  $F$  can also be inferred by  $G$  and every FD that cannot be inferred by  $F$  also cannot be inferred by  $G$
17. Which of the following is not correct?
- A. Every 2-ary relation belongs to 2NF.
- B. Every 2-ary relation belongs to 3NF.
- C. Every 2-ary relation belongs to BCNF.
- D. All the above are not correct.
18. A primary attribute is \_\_\_\_\_.
- A. an attribute in the primary key      B. an attribute not in any candidate key
- C. an attribute in a candidate key      D. an attribute not in the above three situations
19. Given relation schema  $R(ABCDEFGHIJKLMNO PQ)$  and its FD set  $F = \{ AG \rightarrow B, DE \rightarrow C, CHIJ \rightarrow AG \}$ , then  $R'$ 's candidate key is \_\_\_\_\_.
- A. AGDIJKLMNO PQ      B. DECHIJKMNO PQ
- C. DEHIJKLMNO PQ      D. AECHIJKMNO PQ
20. One disadvantage of decomposing relation schema is that \_\_\_\_\_.
- A. data cannot be processed in a centralized way
- B. we may lose some data
- C. we may encounter operation anomalies

D. the time-consuming join operation is often needed in most queries

## Chapter 18~19

For each of the following questions there are four choices marked A, B, C and D. Only one of them is correct. You should decide on the correct choice and write down its mark on the answer sheet.

1. The view mechanism of SQL can enhance the \_\_\_\_\_ of database systems.  
A. integrity      B. concurrency control      C. isolation      D. security
2. The GRANT and REVOKE of SQL can be used to maintain the \_\_\_\_\_ of database systems.  
A. integrity      B. reliability      C. security      D. consistency
3. The security verification can prevent \_\_\_\_\_.  
A. incorrect data to be entered into database  
B. unauthorized users and operations to access database  
C. inconsistent data to be entered into database  
D. authorized users and operations to copy data from databases
4. The integrity verification can prevent \_\_\_\_\_.  
A. incorrect data to be entered into database  
B. unauthorized users to access database  
C. unauthorized operations to access database  
D. correct data to be deleted from database
5. In the following SQL commands, \_\_\_\_\_ belongs to data control commands.  
A. GRANT      B. DROP      C. UPDATE      D. ALTER
6. In order to define primary keys using SQL, we should use \_\_\_\_\_ clause.  
A. PRIMARY KEY      B. GRANT      C. REFERENCES      D. ALTER
7. In order to define foreign keys using SQL, we should use \_\_\_\_\_ clause.  
A. DROP      B. GRANT      C. REFERENCES      D. ALTER
8. The \_\_\_\_\_ of transactions has "all or nothing" property.  
A. atomicity      B. consistency      C. isolation      D. durability
9. The \_\_\_\_\_ of transactions means that partial effects of incomplete transactions should not be visible to other transactions.  
A. atomicity      B. consistency      C. isolation      D. durability
10. The \_\_\_\_\_ of transactions means that effects of a committed transaction are permanent and must not be lost because of later failure.  
A. atomicity      B. consistency      C. isolation      D. durability
11. The \_\_\_\_\_ of transactions means that the execution of any transaction must transform database from one consistent state to another.  
A. atomicity      B. consistency      C. isolation      D. durability
12. In SQL, the function of COMMIT statement is \_\_\_\_\_.  
A. finishing a program      B. returning to DBMS  
C. finishing a transaction successfully      D. storing data
13. In SQL, the function of \_\_\_\_\_ statement is canceling the effects of a transaction to database.  
A. CREATE      B. ROLLBACK      C. DROP      D. COMMIT
14. \_\_\_\_\_ is the most widely used approach to ensure serializability.

- A. Locking      B. Access control      C. Database restoring      D. Consultation
15. Which of the following is not the potential problems caused by concurrency?
- A. Lost update problem      B. Uncommitted dependency problem  
C. Dead lock problem      D. Inconsistent analysis problem
16. DBMS usually adopt \_\_\_\_\_ approach to ensure the correctness of a schedule.
- A. indexing      B. authorization      C. Locking      D. logging
17. A transaction follows \_\_\_\_ if all locking operations precede the first unlock operation in the transaction.
- A. 2PL protocol      B. 2LP protocol      C. 3PL protocol      D. 2LP protocol
18. If transaction has exclusive lock on an item, then it can \_\_\_\_\_ the item.
- A. read but not update      B. update but not read  
C. both read and update      D. neither read nor update
19. When transaction T1 and T2 simultaneously operate data object X, then there are the following four possible situations, where there will be no conflicts in situation \_\_\_\_\_.
- A. while T1 is reading X, T2 wants to write X  
B. while T1 is writing X, T2 wants to write X  
C. while T1 is writing X, T2 wants to read X  
D. while T1 is reading X, T2 wants to read X
20. Suppose there are two transactions that operate the same data, if \_\_\_\_\_, then there will be no conflicts between each other.
- A. one is DELETE, the other is SELECT      B. one is SELECT, the other is UPDATE  
C. both are UPDATE      D. both are SELECT
21. A situation that may result when two (or more) transactions are each waiting for locks held by the other to be released is called \_\_\_\_\_.
- A. starvation      B. deadlock      C. exclusive lock      D. shared lock
22. Which of the following method will never cause deadlocks?
- A. timestamping method      B. timeouts  
C. deadlock detection and recovery      D. two-phase locking protocol
23. In 2PL, if we leave the release of all locks until end of each transaction, then \_\_\_\_\_ can be avoided.
- A. cascading rollback      B. lock contention      C. lost update problem  
D. deadlock
24. Situations that cause the loss of data on secondary storage due to the failure of some hardware during the system running process are called \_\_\_\_\_.
- A. media failures      B. running problems      C. system crashes      D. software errors
25. Logging facilities are used to save \_\_\_\_\_.
- A. the entire process of running a program      B. the result of running a program  
C. update operations to database      D. query operations to database
26. A transaction is a \_\_\_\_\_.
- A. program      B. procedure      C. sequence of operations      D. integrity rule
27. In order to recover a database, the important clue is \_\_\_\_\_.
- A. DBA      B. DD      C. document      D. log file.
28. The main aim of backup mechanism is \_\_\_\_\_.
- A. to control concurrency      B. to save historical archive

- C. to recover database from failures                      D. to control security
29. If failure occurs between commit and flushing database buffers to secondary storage, then recovery manager has to \_\_\_\_ the transaction.  
A. REDO                      B. UNDO                      C. ABORT                      D. ROLLBACK
30. If transaction had not committed at failure time, recovery manager has to \_\_\_\_ the transaction.  
A. REDO                      B. UNDO                      C. ABORT                      D. ROLLFORWARD
31. In database technology, "dirty data" refers to the data \_\_\_\_\_.  
A. that are not rolled back                      B. that are not committed  
C. that are rolled back                      D. that are not committed but rolled back subsequently
32. A sequence of reads/writes in concurrent transactions is called a \_\_\_\_\_.  
A. procedure                      B. step                      C. schedule                      D. priority
33. \_\_\_\_\_ if and only if wait-for graph contains a cycle.  
A. Starvation happens                      B. Deadlock exists  
C. Execution succeeds                      D. Execution fails

## Chapter \*: Database Design

- A. For each of the following questions there are four choices marked A, B, C and D. Only one of them is correct. You should decide on the correct choice and write down its mark on the answer sheet.**
1. DFD is the description tool used in the \_\_\_\_\_ phase of database design.  
A. outline design                      B. feasibility analysis  
C. program coding                      D. requirement analysis
2. During database design process, the task that transforms an E-R diagram into a set of relation schemas belongs to the \_\_\_\_\_ phase.  
A. requirement analysis                      B. logical structure design  
C. conceptual structure design                      D. physical structure design
3. The common method and description tool of expressing the conceptual structure of a database are \_\_\_\_\_.  
A. hierarchy analysis and hierarchy structure diagram  
B. data flow analysis and data flow diagram  
C. entity relationship analysis and E-R diagram  
D. structure analysis and module structure diagram
4. The design of relation schemas in designing relational database is the task of \_\_\_\_\_ phrase.  
A. logical structure design                      B. conceptual structure design  
C. physical structure design                      D. requirement analysis
5. The main problem that involves normalization theory of relational database is how to \_\_\_\_\_.  
A. construct proper logical structure to reduce data redundancy and operation anomaly  
B. construct proper physical structure to enhance the running efficiency of database  
C. construct proper application interface for the user to conveniently use database  
D. control operational privileges so that the unauthorized access to database can be prevented
6. Database design process can be divided into 6 phases. Each phase has its own design actions.

“ the determination of creating what indexes on what attributes of what relations ” is a task that should belong to \_\_\_\_\_ phase.

- A. requirement analysis      B. conceptual structure design
  - C. logical structure design      D. physical structure design
7. In deriving an relational model from an E-R diagram, if the relationship between two entities is  $m:n$  relationship, then the correct transforming method is \_\_\_\_\_ .
- A. adding the primary key of the entity at  $n$  side and the attributes on the relationship into  $m$  side
  - B. adding the primary key of the entity at  $m$  side and the attributes on the relationship into  $n$  side
  - C. representing the relationship using a new relation and adding the primary keys of the entities at both  $m$  side and  $n$  side into it
  - D. adding into the entity at  $m$  side an attribute that represents rank of record.
8. Suppose we are given an E-R diagram that consists of 3 entities and 3  $m:n$  relationships, then according to the rules of transforming E-R diagram into relational model, the result relational model should include \_\_\_\_\_ relation schemas.
- A. 4      B. 5      C. 6      D. 7
9. During requirement analysis phase, we often use \_\_\_\_\_ method to design DFD.
- A. object oriented      B. trace back      C. bottom up      D. top down
10. During conceptual structure design phase, we often use \_\_\_\_\_ method to design conceptual structure.
- A. object oriented      B. trace back      C. bottom up      D. top down
11. Conceptual structure should meet the needs of \_\_\_\_\_.
- A. the programming request of application programmers
  - B. the information management request of DBAs
  - C. the maintenance request of database systems
  - D. the information request of organizations
12. The bridge of communicating between database designers and users is \_\_\_\_\_.
- A. program flow chart      B. entity-relationship diagram
  - C. module structure diagram      D. data structure diagram
13. In transforming E-R model into relational model, among the following descriptions, the wrong one is \_\_\_\_\_.
- A. each entity type should be transformed into a relation schema
  - B. each relationship type should be transformed into a relation schema
  - C. each  $m:n$  relationship type should be transformed into a relation schema
  - D. generally, 1:1 and 1: $n$  relationship types are not transformed into relation schemas
14. If there is a 1: $n$  relationship type between entities from the same entity set, then according to the rules of transforming E-R diagram into relational model, the result relational model should include \_\_\_\_\_ relation schemas.
- A. 1      B. 2      C. 3      D. 4
15. If there is a  $m:n$  relationship type between entities from the same entity set, then according to the rules of transforming E-R diagram into relational model, the result relational model should include \_\_\_\_\_ relation schemas.
- A. 1      B. 2      C. 3      D. 4

## B. Database design

Please design a database to manage courses, teachers, students and their relationships based on the following description.

A teacher can teach zero or more courses; a course can be taught by zero or more teachers; a student can take part in zero or more courses; a course can be learnt by zero or more students. For a student taking part in a course, there should be a grade; for a teacher teaching a course, there should be a workload. Courses should have attributes such as course number, course name and credit hour; teachers should have attributes such as teacher number, teacher name and teacher sex; students should have attributes such as student number, student name and student sex.

- (1) Draw corresponding ER diagram.
- (2) Derive relation data model from your ER diagram.
- (3) Give SQL statements to create table in your data model such that: 1) primary key and foreign keys of each table are properly defined; 2) student name, teacher name and sex cannot be null; 3) sex can only take value ' M' or ' F' .

# Answers

## Chapter 1~2

1. C 2. A 3. D 4. C 5. C 6. D 7. A 8. A 9. C 10. A  
11. C 12. D 13. D 14. C 15. A 16. C 17. A 18. B 19. A 20. C  
21. D 22. B 23. C 24. C 25. D 26. B 27. D 28. B 29. A 30. B

## Chapter 3~4

1. A 2. D 3. C 4. C 5. A 6. B 7. D 8. A 9. B 10. C  
11. D 12. C 13. B 14. A 15. B 16. D 17. A 18. B 19. C 20. C  
21. D 22. D 23. B 24. A 25. D 26. B 27. B 28. B

## Chapter 5~6

1. D 2. B 3. C 4. B 5. A 6. D 7. C 8. C 9. B 10. A  
11. D 12. A 13. D 14. A 15. B 16. D 17. A 18. B 19. C 20. C  
21. D 22. D 23. D 24. D 25. D 26. C 27. C

## Chapter 13

1. B 2. B 3. B 4. C 5. B 6. C 7. D 8. C 9. A 10. B  
11. B 12. B 13. B 14. C 15. B 16. D 17. D 18. C 19. C 20. D

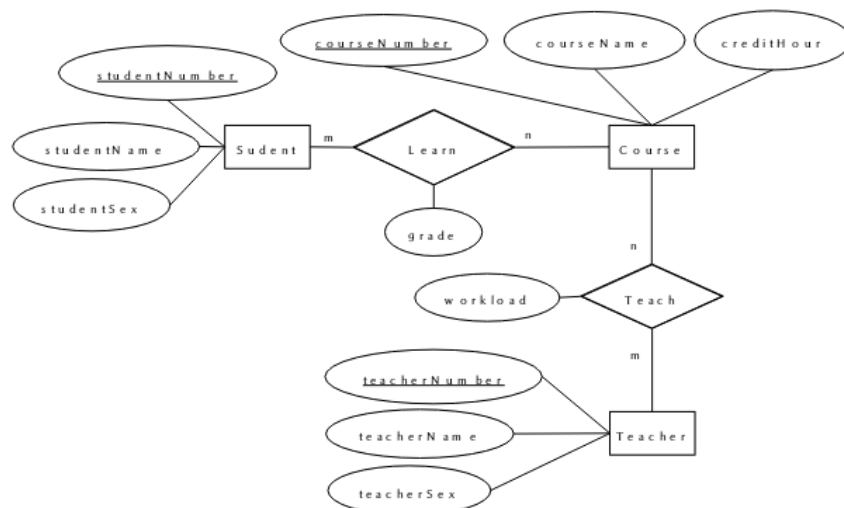
## Chapter 18~19

1. D 2. C 3. B 4. A 5. A 6. A 7. C 8. A 9. C 10. D  
11. B 12. C 13. B 14. A 15. C 16. C 17. A 18. C 19. D 20. D  
21. B 22. A 23. A 24. A 25. C 26. C 27. D 28. C 29. A 30. B  
31. D 32. C 33. B

## Chapter \*: Database Design

- A. 1. D 2. B 3. C 4. A 5. A 6. D 7. C 8. C 9. D 10. C  
11. D 12. B 13. B 14. A 15. B

- B. (1)



(2)

Student (studentNumber, studentName, studentSex )

Primary Key: studentNumber

Course (courseNumber, courseName, creditHour)

Primary Key: courseNumber

Teacher (teacherNumber, teacherName, teacherSex)

Primary Key: teacherNumber

Learning (studentNumber, courseNumber, grade)

Primary Key: (studentNumber, courseNumber)

Foreign key: studentNumber, courseNumber

Teaching (teacherNumber, courseNumber, workload)

Primary Key: (teacherNumber, courseNumber)

Foreign key: teacherNumber, courseNumber

(3)

create table Student (

studentNumber char(5) primary key,

studentName char(10) not null,

studentSex char(1) not null check (studentSex IN ( ' M' , ' F' ) )

)

create table Course (

courseNumber char(5) primary key,

courseName char(40),

creditHour INT

)

create table Teacher (

teacherNumber char(5) primary key,

teacherName char(10) not null,

teacherSex char(1) not null check (teacherSex IN ( ' M' , ' F' ) )

)

create table Learning (

studentNumber char(5) foreign key references Student(studentNumber),

courseNumber char(5) foreign key references Course(courseNumber),

grade float,

primary key (studentNumber, courseNumber)

)

create table Teaching (

teacherNumber char(5) foreign key references Teacher(teacherNumber),

courseNumber char(5) foreign key references Course(courseNumber),

workload float,

primary key (teacherNumber, courseNumber)

)